Experiment No. :- 4

THEORY

1. Data Comparison :- Used to Compare joining data using <, > etc.

2. LENGTH() :- Return no. of Characters in String.

3. UPDATE :- Modifies existing records in a table.

4. Percentage Calculations :- Used for Salary increment (eg - 10% increase)

5. Pattern with :- Checks Specific Character position in a String.

```
MariaDB [vijendra0862]> SELECT * FROM employee WHERE hiredate < "1980-06-30" OR hiredate > "1981-12-30";
+-------+-------+-------+------+------------+------+------+--------+
| empno | ename | job   | mgr  | hiredate   | sal  | comm | deptno |
+-------+-------+-------+------+------------+------+------+--------+
| 7369  | SCOTT | ANALYST | 7566 | 1982-12-09 | 3000 | NULL |     40 |
| 7876  | ADAMS | CLERK | 7788 | 1983-01-12 | 1100 | NULL |     20 |
| 7934  | MILLER | CLERK | 7782 | 1982-01-23 | 1300 | NULL |     10 |
+-------+-------+-------+------+------------+------+------+--------+
3 rows in set (0.000 sec)
```

```
MariaDB [vijendra0862]> SELECT ename FROM employee WHERE ename Like "_A%";
+--------+
| ename  |
+--------+
| WARD   |
| MARTIN |
| JAMES  |
+--------+
3 rows in set (0.001 sec)
```

```
MariaDB [vijendra0862]> SELECT ename FROM employee WHERE ename Like "_____";
+--------+
| ename  |
+--------+
| SMITH  |
| ALLEN  |
| JONES  |
| BLAKE  |
| CLARK  |
| SCOTT  |
| ADAMS  |
| JAMES  |
+--------+
8 rows in set (0.001 sec)
```

```
MariaDB [vijendra0862]> SELECT ename FROM employee WHERE ename Like "_A%";
+--------+
| ename  |
+--------+
| WARD   |
| MARTIN |
| JAMES  |
+--------+
3 rows in set (0.001 sec)
```

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4 1. Display the list of employees who have joined the Company before 30th June 80 or after 31st Dec 81.

MariaDB [Vijendra0862] > SELECT employee WHERE hiredate < "1980-06-30" OR hiredate > "1981-12-30";

2 Display the names of employees whose names have Second alphabet A in their names

MariaDB [Vijendra0862] > SELECT ename FROM employee WHERE ename Like "_A%";

3 Display the names of employees whose name is exactly five characters in length

MariaDB [Vijendra0862] > SELECT ename FROM employee WHERE ename LIKE '_____';

4 Display the names of employees whose names have Second alphabet A in their name

MariaDB [Vijendra0862] > SELECT ename FROM employee WHERE ename LIKE '_A%';

TRYON

Teacher's Sign.

```
MariaDB [vijendra0862]>
MariaDB [vijendra0862]> SELECT ename
    -> FROM employee
    -> WHERE job NOT IN ('SALESMAN', 'CLERK', 'ANALYST');
+-------+
| ename |
+-------+
| JONES |
| BLAKE |
| CLARK |
| KING  |
+-------+
4 rows in set (0.002 sec)
```

```
MariaDB [vijendra0862]> SELECT ename, sal*12 AS annual_sal FROM employee order by annual_sal desc;
+--------+------------+
| ename  | annual_sal |
+--------+------------+
| KING   |      60000 |
| SCOTT  |      36000 |
| FORD   |      36000 |
| JONES  |      35700 |
| BLAKE  |      34200 |
| CLARK  |      29400 |
| ALLEN  |      19200 |
| TURNER |      18000 |
| MILLER |      15600 |
| MARTIN |      15000 |
| WARD   |      15000 |
| ADAMS  |      13200 |
| JAMES  |      11400 |
| SMITH  |       9600 |
+--------+------------+
14 rows in set (0.001 sec)
```

```
MariaDB [vijendra0862]> SELECT
    -> ename,
    -> sal,
    -> sal*0.15 AS hra,
    -> sal*0.10 AS da,
    -> sal*0.05 AS pf,
    -> (sal + sal*0.15 + sal*0.10 - sal*0.05) AS totalsal
    -> FROM employee
    -> ORDER BY totalsal, hra;
+--------+------+--------+--------+--------+----------+
| ename  | sal  | hra    | da     | pf     | totalsal |
+--------+------+--------+--------+--------+----------+
| SMITH  |  800 | 120.00 |  80.00 |  40.00 |   960.00 |
| JAMES  |  950 | 142.50 |  95.00 |  47.50 |  1140.00 |
| ADAMS  | 1100 | 165.00 | 110.00 |  55.00 |  1320.00 |
| WARD   | 1250 | 187.50 | 125.00 |  62.50 |  1500.00 |
| MARTIN | 1250 | 187.50 | 125.00 |  62.50 |  1500.00 |
| MILLER | 1300 | 195.00 | 130.00 |  65.00 |  1560.00 |
| TURNER | 1500 | 225.00 | 150.00 |  75.00 |  1800.00 |
| ALLEN  | 1600 | 240.00 | 160.00 |  80.00 |  1920.00 |
| CLARK  | 2450 | 367.50 | 245.00 | 122.50 |  2940.00 |
| BLAKE  | 2850 | 427.50 | 285.00 | 142.50 |  3420.00 |
| JONES  | 2975 | 446.25 | 297.50 | 148.75 |  3570.00 |
| FORD   | 3000 | 450.00 | 300.00 | 150.00 |  3600.00 |
| SCOTT  | 3000 | 450.00 | 300.00 | 150.00 |  3600.00 |
| KING   | 5000 | 750.00 | 500.00 | 250.00 |  6000.00 |
+--------+------+--------+--------+--------+----------+
14 rows in set (0.001 sec)
```

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Topic :

5. Display the names of empolyees who are not working as Salesman or Clerk or analyst.

MariaDB [Vijendra0862]> SELECT ename FROM employee WHERE job NOT IN('SALESMAN', 'CLERK', 'ANALYST')

6. Display the name of the employee along with their annual Salary (Sal*12). The name of the employee earning highest Salary should appear first.

MariaDB [Vijendra0862]> SELECT ename, Sal*12 AS annual salary FROM employee ORDER By Sal DESC;

7. Display name, Sal, hra, Pf, da, totalsal for each employee. The Output Should be in the order of total Sal, hra 15% of Sal, da 10% of Sal, pf 5% of Sal. Total Sal will be (Sal+hra+da)-Pf.

MariaDB [Vijendra0862]> SELECT ename, Sal, Sal*0.15 As hra, Sal*0.10 As da, Sal*0.05 AS pf, (Sal+Sal*0.15 + Sal*0.10 - Sal*0.05) AS totalsal FROM employee ORDER By totalsal, hra;

TRYON

Teacher's Sign.

```
MariaDB [vijendra0862]> SELECT ename, sal, sal*1.10 AS new_salary
    -> FROM employee
    -> WHERE comm IS NULL;
+-------+------+------------+
| ename | sal  | new_salary |
+-------+------+------------+
| SMITH |  880 |     968.00 |
| JONES | 3273 |    3600.30 |
| BLAKE | 3135 |    3448.50 |
| CLARK | 2695 |    2964.50 |
| SCOTT | 3300 |    3630.00 |
| KING  | 5500 |    6050.00 |
| ADAMS | 1210 |    1331.00 |
| JAMES | 1045 |    1149.50 |
| FORD  | 3300 |    3630.00 |
| MILLER | 1430 |    1573.00 |
+-------+------+------------+
10 rows in set (0.001 sec)
```

```
MariaDB [vijendra0862]> SELECT ename, sal FROM employee WHERE sal*1.20 > 3000;
+-------+------+
| ename | sal  |
+-------+------+
| JONES | 3273 |
| BLAKE | 3135 |
| CLARK | 2695 |
| SCOTT | 3300 |
| KING  | 5500 |
| FORD  | 3300 |
+-------+------+
6 rows in set (0.002 sec)
```

```
MariaDB [vijendra0862]> SELECT ename, sal FROM employee WHERE sal > 100;
+-------+------+
| ename | sal  |
+-------+------+
| SMITH |  880 |
| ALLEN | 1600 |
| WARD  | 1250 |
| JONES | 3273 |
| MARTIN | 1250 |
| BLAKE | 3135 |
| CLARK | 2695 |
| SCOTT | 3300 |
| KING  | 5500 |
| TURNER | 1500 |
| ADAMS | 1210 |
| JAMES | 1045 |
| FORD  | 3300 |
| MILLER | 1430 |
+-------+------+
14 rows in set (0.001 sec)
```

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8 Update the Salary of each employee by 10% increment who are not eligible for Commission.

MariaDB [Vijendra0862]> UPDATE employee SET Sal = Sal*1.10 WHERE Comm IS NULL;

9. Display those employees whose Salary is more than 3000 after giving 20% increment.

MariaDB [Vijendra0862]> SELECT ename, Sal FROM employee WHERE Sal*1.20>3000;

10. Display those employees whose Salary Contains at ~~least~~ 3 digits.

MariaDB [Vijendra0862]> SELECT ename, Sal FROM employee WHERE Sal >= 100;

Vidhar
26/02/26

Teacher's Sign.

TRYON